Gennaio 2024

1. Digital Signature and Digital Certificates

(a) You want to digitally sign a file which is stored in your PC. A smartcard reader is connected to your PC and your smartcard is inserted into the smartcard reader. On a client application running on your PC you issue the request to digitally sign the file. Please fill in the blanks.

1. The smartcard reader may ask the user to type a PIN.
2. Your PC sends the hash code of the file to the smartcard reader.
3. The private key is stored on the smartcard.
4. The smartcard computes the encryption of the hash code and returns the result to the PC.

(b) Assume you are a Server supporting the SSL protocol. Upon request by the client you reply with your digital certificate. At this point you get a string X of bit from the client. What do you do next?

A. I take X as the session key shared with the client.

B. I decrypt X with the private key of the client and take the result as the session key shared with the client.

C. I decrypt X with the public key of the client and take the result as the session key shared with the client.

X I decrypt X with my private key and take the result as the session key shared with the client.

E. I decrypt X with my public key and take the result as the session key shared with the client.

2. Cryptography

Consider a block cipher operating on 4-bit blocks, written as hexadecimal digits.

For one particular key K, it implements the following permutation:

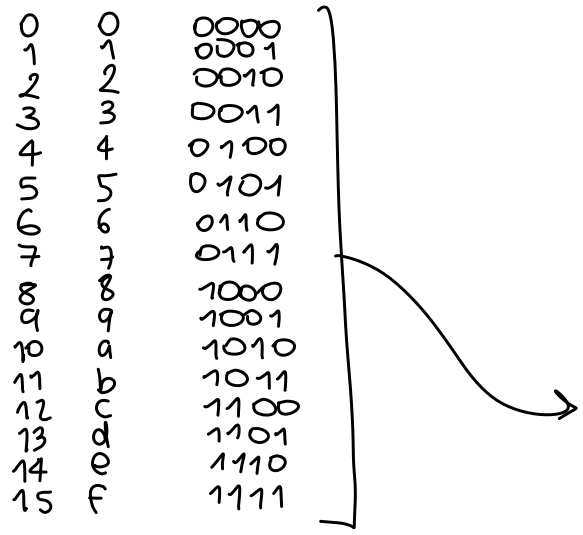
m	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
$E_K(m)$	0	6	5	c	7	e	2	a	4	9	f	d	8	3	b	1

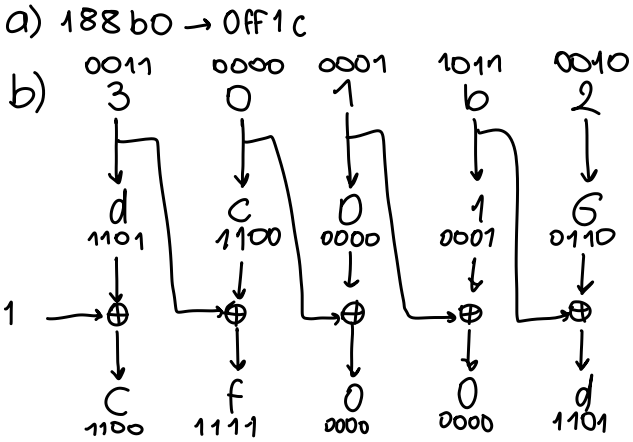
Using this key K, decrypt the following three ciphertexts according to the indicated modes of operation. You may find it useful the following table:

⊕	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
0	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
1	1	0	3	2	5	4	7	6	9	8	b	a	d	c	f	e
2	2	3	0	1	6	7	4	5	a	b	8	9	e	f	c	d
3	3	2	1	0	7	6	5	4	b	a	9	8	f	e	d	c
4	4	5	6	7	0	1	2	3	c	d	e	f	8	9	a	b
5	5	4	7	6	1	0	3	2	d	c	f	e	9	8	b	a
6	6	7	4	5	2	3	0	1	e	f	c	d	a	b	8	9
7	7	6	5	4	3	2	1	0	f	e	d	c	b	a	9	8
8	8	9	a	b	c	d	e	f	0	1	2	3	4	5	6	7
9	9	8	b	a	d	c	f	e	1	0	3	2	5	4	7	6
a	a	b	8	9	e	f	c	d	2	3	0	1	6	7	4	5
b	b	a	9	8	f	e	d	c	3	2	1	0	7	6	5	4
c	c	d	e	f	8	9	a	b	4	5	6	7	0	1	2	3
d	d	c	f	e	9	8	b	a	5	4	7	6	1	0	3	2
e	e	f	c	d	a	b	8	9	6	7	4	5	2	3	0	1
f	f	e	d	c	b	a	9	8	7	6	5	4	3	2	1	0

(a) Electronic Code Book (ECB): 188b0

(b) Cipher-Block Chaining (CBC) – assume $IV = 1$: 301b2





3. Public Key Cryptography

In a chemical plant a sensor reads the temperature T of a boiler and sends it to a controller. Upon receipt of a message with a temperature greater than 100 C, the controller triggers an emergency procedure that halts the plant. Since halting the plan implies considerable financial losses, false alarms must be avoided. To ensure the authenticity of the message, the sensor sends messages of the form $(T, E(K_S^{-1}, T))$ where K_S^{-1} is its private key. The corresponding public key, $K_S = (109, 551)$, is used by the controller to verify the authenticity of the message.

You are an attacker and your aim is to disrupt the production of the chemical plant. You have access to the commnunication channel used by the sensor to send messages to the controller and you have overheard the message $(11, 524)$ (numbers are in decimal notation). You have a **single** opportunity to send a message (T', A') to the controller over this channel.

What values for T' and A' would you choose to achieve your goal?

$T' = 121$

$A' = (524 \cdot 524) \bmod 551$

To avoid complex calculations, you can write expressions containing numbers (no variables) possibly combined with arithmetic operators (no exponantiation). For example:

- valid expressions: $123, 123 + 456, 123 * 456$
- invalid expressionss: $123 * A, 123^{456}$

Please justify your answer in the following box.

$K_S = (109, 551)$
$n = 551$
$e = 109$
messaggio : $(11, 524)$

$firma(T_1 \cdot T_2) = (A_1 \cdot A_2) \bmod n$
temperatura $> 100 \rightarrow 11 \cdot 11 = 121$

$A' = A \cdot A \bmod n$
$A' = 524 \cdot 524 \bmod 551$

4. Security Protocols

The following key exchange protocol has been designed to allow A to guess and share a new (i.e. fresh) session key K with B. Server S is is assumed to be trustworthy and the publioc key of S, namely K_S, is the only public key that A knows in advance.

$1. A \rightarrow S : A, B$
$2. A \leftarrow S : E(K_S^{-1}, [A, K_A]), E(K_S^{-1}, [B, K_B])$
$3. A \rightarrow B : E(K_B, [K, N_A])$
$4. A \leftarrow B : E(K, [K, N_A])$

where N_A is a nonce.

(a) Show that the protocol does not meet its design objetives by presenting an attack.

(b) Show how would you fix the procol by modifying step 3.